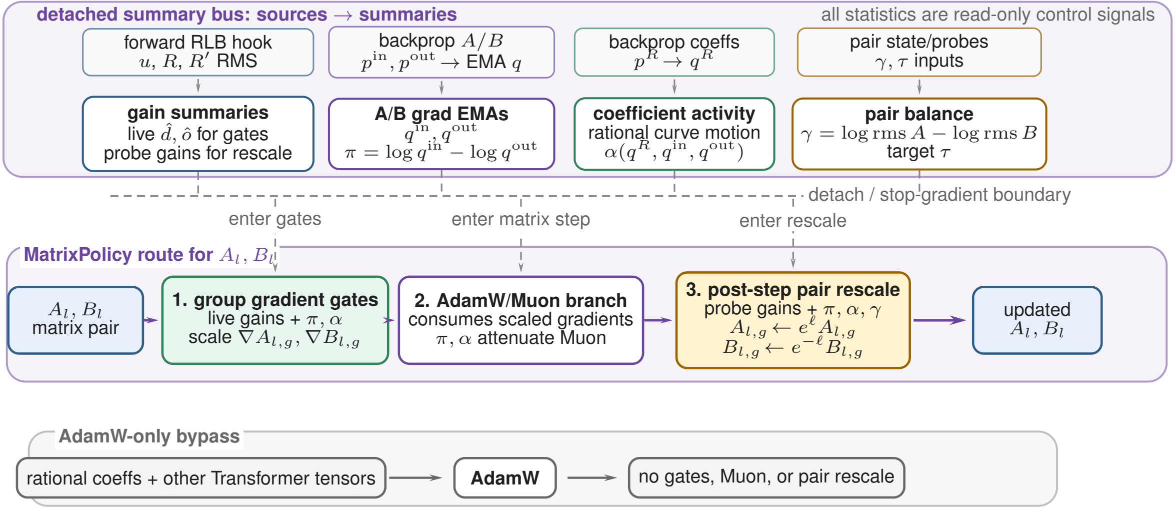

detached summary bus: sources → summaries
all statistics are read-only control signals
forward RLB hook
u, R, R' RMS
backprop A/B
$p^{\text{in}}, p^{\text{out}} \to$ EMA q
backprop coeffs
$p^R \to q^R$
pair state/probes
γ, τ inputs
gain summaries
live $\hat{d}$, $\hat{o}$ for gates
probe gains for rescale
A/B grad EMAs
$q^{\text{in}}, q^{\text{out}}$
$\pi = \log q^{\text{in}} - \log q^{\text{out}}$
coefficient activity
rational curve motion
$\alpha(q^R, q^{\text{in}}, q^{\text{out}})$
pair balance
$\gamma = \log \text{rms}\, A - \log \text{rms}\, B$
target τ
detach / stop-gradient boundary
enter gates
enter matrix step
enter rescale
MatrixPolicy route for A_l, B_l
A_l, B_l
matrix pair
1. group gradient gates
live gains + π, α
scale $\nabla A_{l,g}, \nabla B_{l,g}$
2. AdamW/Muon branch
consumes scaled gradients
π, α attenuate Muon
3. post-step pair rescale
probe gains + π, α, γ
$A_{l,g} \leftarrow e^{\ell} A_{l,g}$
$B_{l,g} \leftarrow e^{-\ell} B_{l,g}$
updated
A_l, B_l
AdamW-only bypass
rational coeffs + other Transformer tensors
AdamW
no gates, Muon, or pair rescale